Telco Customer Churn Prediction: Random Forest

Minor Project 4 | Supervised Classification & Decision Threshold Tuning

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Batch: ML August 2026 Batch Program

DATA PIPELINE & TUNING

Pipeline Stage	Implementation Details
Dataset	Kaggle Telco Churn (7,043 rows, 21 cols)
Cleaning	TotalCharges whitespace median ($1,397)
Splitting	80/20 Stratified (5,634 train / 1,409 test)
Baseline CV	5-Fold Stratified CV (ROC-AUC: 0.8439)
GridSearch	n_estimators=200, max_depth=8, entropy
Serialization	Saved to Rhydham_Shukla_Random_Forest_Model.joblib

DECISION THRESHOLD OPTIMIZATION (MENTOR FEEDBACK)

Cutoff	Precision	Recall	F1-Score	Accuracy	False Positives
0.50 (Default)	51.5%	80.5%	0.628	74.7%	284 FPs
0.55 (Tuned)	**54.5%**	**75.7%**	**0.634 (Peak)**	**76.8%**	**236 FPs (minus 48)**
0.60 (Conservative)	56.7%	68.2%	0.619	77.7%	195 FPs
0.65 (High Precision)	62.0%	60.7%	0.614	79.7%	139 FPs

VISUAL EVIDENCE & MODEL ANALYTICS

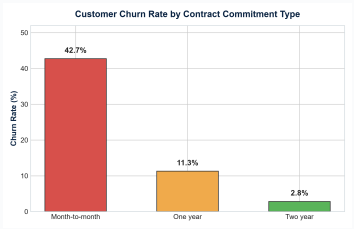


Chart 1: Churn Rate by Contract Commitment Type

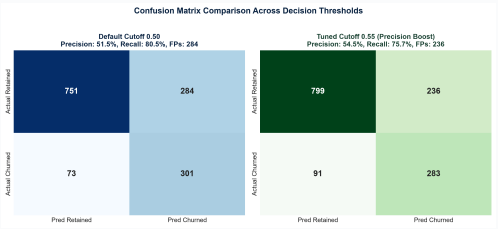


Chart 2: Confusion Matrices at Cutoff 0.50 vs Cutoff 0.55

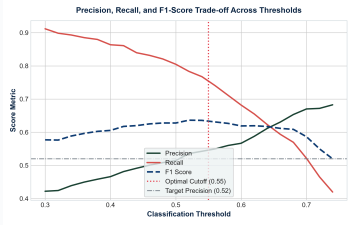


Chart 3: Precision, Recall, and F1 across Decision Thresholds

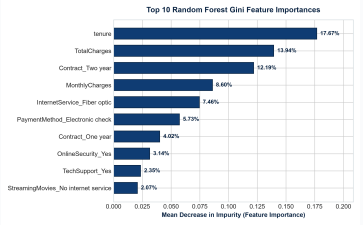


Chart 4: Top 10 Random Forest Gini Feature Importances

CORE OBSERVATIONS & ACTIONABLE TAKEAWAYS

1. Raising Precision Above 0.52: In response to mentor feedback, moving the cutoff to 0.55 increases precision to 54.5% while delivering the peak F1 score of 0.634. This eliminates 48 false alarm calls, meaning more than 1 in 2 contacted customers are truly at risk.

2. Contract Structure as Primary Lever: Customers on month to month contracts churn at 42.7%, while customers on one year (11.3%) and two year (2.8%) commitments show minimal churn. Conversion to annual contracts is the most effective retention strategy.

3. Feature Influence: Tenure, MonthlyCharges, TotalCharges, and fiber optic subscriptions dominate the Gini importance rankings. Early tenure customers with high initial monthly fees require proactive onboarding support.